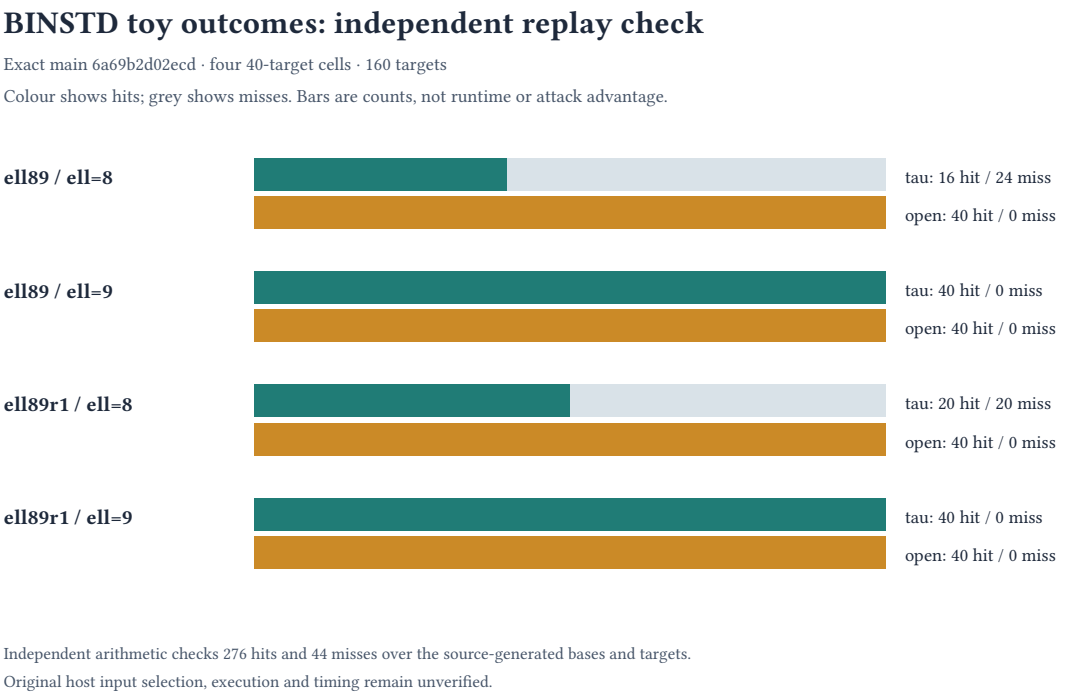
BINSTD archived toy outcome audit

26 October 2026 · crypto-autoresearcher main `6a69b2d02ecd53f1649bcf56e23e4ffe91d953ad`

This is a source-bound audit of four archived `n=17`, `ell=8/9` cells. It adds no run or timing measurement.

Inputs and independent check

The four frozen sheriff per-target receipts name 160 targets, 276 positive hit witnesses and 44 misses. `check.py` rehashes six source/archive paths per receipt at their historical commits; it checks receipt, raw-result and yield-matrix bindings. It verifies all targets and witness triples on the curve `y²+xy=x³+1` using independent interleaved field multiplication and Euclid inversion, and rejects an on-curve witness mutation in each cell. The addition formula is checked against SEC 1 v2 §2.2.2. It matches all seeded targets exactly, and factor-base sizes and pair-sum cardinalities to the receipts, then independently enumerates all base-pair sums and target-minus-base-point lookups. Each hit and miss flag agrees; a deliberately flipped false hit is rejected where misses exist.



Observation

Cell	Tau hit/miss	Open hit/miss	Witnesses
ell89 / ell 8	16/24	40/0	56
ell89 / ell 9	40/0	40/0	80
ell89r1 / ell 8	20/20	40/0	60
ell89r1 / ell 9	40/0	40/0	80

All archived count pairs satisfy the premises of the **conditional** Lean `YieldCeiling.saturated_open_below_band` theorem. Exhaustive finite trial division independently checks that the archived `t^17+t^3+1` modulus and degree-eight factors are irreducible, and their product is `X^17+1`. This is computed evidence. A new Lean theorem also kernel-checks the 510/30/30 exact **bitmask** divisor trials using `decide`; its axiom audit lists only `propext`. The same predicate rejects encoded reducible `X^17+1` and `X^8+1` controls, with degree-one witness `X+1`. It does not yet prove polynomial irreducibility. A separate set-polynomial Rabin test obtains Frobenius residue `X` and GCD `1` for all three masks; reducible `X^17+1` and `X^8+1` controls fail.

Boundary and next check

The finite exhaustive check covers each reported miss for the **source-generated** factor bases and targets. It does not establish the original host's input selection, per-target execution or timing. The normal-basis/Frobenius bridge is also not formalized. Obtain the original trace or execution proof; then prove that the encoded remainder agrees with `(ZMod 2)[X]`, identify the archived masks with the polynomial factors, and prove the Frobenius bridge.

Source: `research/binstd_end_to_end_20261026/REPORT.md`, `evidence.json`, `formal/CryptoResearch/Binstd/EncodedDivisors.lean`, and SEC 1 v2 §2.2.2 (https://www.secg.org/sec1-v2.pdf). Rabin's criterion: https://doi.org/10.1137/0209024.